FURTHER MECHANICS / WORK, ENERGY AND POWER

Elastic Potential Energy

Questions by topic 10 questions · 101 marks

Instructions

Unless a question states otherwise, take $g = 9.8\,\mathrm{m\,s^{-2}}$ wherever a numerical value of g is required. Solutions also give final answers for $g = 10\,\mathrm{m\,s^{-2}}$ where these differ.

Questions

Question 1	*6 marks*	2
Question 2	*11 marks*	3
Question 3	*5 marks*	4
Question 4	*9 marks*	5
Question 5	*14 marks*	6
Question 6	*11 marks*	7
Question 7	*12 marks*	8
Question 8	*11 marks*	9
Question 9	*11 marks*	10
Question 10	*11 marks*	11

1. One end of a light elastic string, of natural length $2a$ and modulus of elasticity $2mg$, is attached to a fixed point O. The other end of the string is attached to a particle P of mass m. The string is vertical with P directly above O. The particle P is pulled vertically upwards until the extension of the string is $2a$. The particle P is then released from rest.

(a) Find the speed of P when it is at a distance $\frac{3}{2}a$ above O. **[4]**

(b) Find the initial acceleration of P when it is released from rest. **[2]**

2. A light elastic string has natural length 2.5 m and modulus of elasticity 15 N.

One end of the elastic string is attached to a particle of mass 0.75 kg.

The other end of the elastic string is attached to a fixed point O.

The particle is released from rest at a point A, which is 2.9 m vertically below O.

(a) Calculate the elastic potential energy stored in the string when the particle is at A. **[2]**

(b) The point B is 4.0 m vertically below O.
Calculate the gravitational potential energy lost by the particle as it moves from A to B. **[2]**

(c) Find the speed of the particle at B. **[4]**

(d) The point C is 3.3 m vertically below O.
Explain, showing any calculations that you make, why the speed of the particle is increasing the first time that the particle is at C. **[3]**

3. A particle P of mass m is attached to one end of a light elastic string of natural length a and modulus $2mg$. The other end of the string is attached to a fixed point O on a rough plane inclined at $30°$ to the horizontal.

The string lies along a line of greatest slope, with O below P. The particle is initially at rest on the plane with the string at its natural length.

The coefficient of friction between P and the plane is $\frac{1}{\sqrt{3}}$.

The particle is projected up the plane, away from O, with speed $\sqrt{\frac{3ga}{2}}$.

Find the greatest extension of the string during the subsequent motion. **[5]**

4. The points A and B are at the same horizontal level and are $24a$ apart. The ends of a light elastic string, of natural length $18a$ and modulus of elasticity λ, are attached to A and B. A particle P of mass m is attached to the midpoint of the string. The system is in equilibrium with P at a distance $5a$ below M, the midpoint of AB.

(a) Find λ in terms of m and g. **[3]**

The particle P is pulled down vertically and released from rest from a position $9a$ below M.

(b) Find, in terms of a and g, the speed of P as it passes through its equilibrium position. **[6]**

5. A light elastic string has natural length $3a$ and modulus of elasticity $9mg$.
One end of the string is attached to a fixed point O. A particle P of mass m is attached to the other end.
The particle hangs freely in equilibrium at the point E, vertically below O.

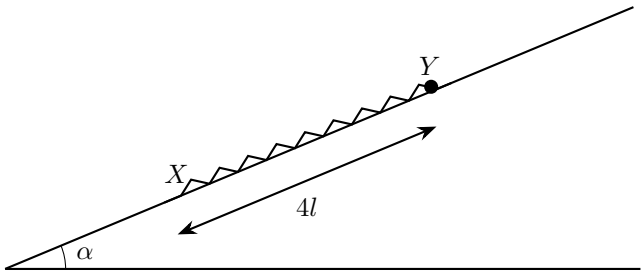
(a) Find the length OE. **[4]**

The particle is then pulled vertically downwards from E to a point A, where $OA = \frac{16a}{3}$, and released from rest.
The resistance to the motion of P is a constant force of magnitude $\frac{1}{4}mg$.

(b) Find, in terms of a and g, the speed of P as it passes through E. **[6]**

The particle is now held at O and released from rest.
The resistance to the motion of P is again a constant force of magnitude $\frac{1}{4}mg$.
The speed of P is greatest when it is at the point B.

(c) Find the distance OB. **[4]**



6. A light elastic spring has natural length $2l$ and modulus of elasticity mg. One end of the spring is attached to a fixed point X on a rough inclined plane. The other end is attached to a package P of mass m.

The plane is inclined to the horizontal at an angle α where

$$\tan\alpha = \frac{5}{12}$$

The package is initially held at the point Y on the plane, where $XY = 4l$. The point Y is higher than X and XY is a line of greatest slope of the plane, as shown. The package is released from rest at Y and moves down the plane.

The coefficient of friction between P and the plane is $\frac{1}{4}$.

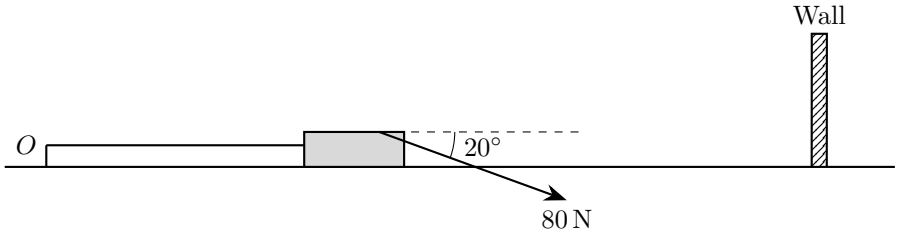
By modelling P as a particle,

(a) show that the acceleration of P at the instant when it is released is

$$\frac{15}{13}g$$

down the plane. **[5]**

(b) find, in terms of g and l, the speed of P at the instant when the spring first regains its natural length. **[6]**



7. A block of mass 6 kg rests on a rough horizontal surface and is attached to a fixed point O by a light elastic string. Initially the block is at rest 2.4 m from O.

The elastic string has natural length 2 m and modulus of elasticity 80 N.

The coefficient of friction between the block and the surface is 0.25.
A force of magnitude 80 N is applied to the block towards a vertical wall. The force acts at an angle of 20° below the horizontal, as shown in the diagram.

The block moves in a straight line perpendicular to the wall, and the wall is sufficiently far away that the block does not reach it during the motion considered.
In this question, take $g = 9.8\,\text{m}\,\text{s}^{-2}$.

(a) Show that the block has moved 0.94 m, correct to 2 significant figures, when the forces acting on it are in equilibrium. **[4]**

(b) State one limitation of the model. **[1]**

(c) Find the maximum speed of the block. **[7]**

8. A spring of natural length a is suspended from a fixed point A. A package P of mass m is attached to the lower end of the spring.

The package is initially at the point B, vertically below A, where $AB = a$.

The package is projected vertically downwards from B with speed $\sqrt{ag}$.

It first comes to instantaneous rest at the point C, where $AC = 2a$.

Air resistance is modelled as negligible. By modelling the spring as light and modelling P as a particle,

(a) show that the modulus of elasticity of the spring is $3mg$. **[4]**

(b) (i) Show that P attains its maximum speed when the extension of the spring is $\frac{1}{3}a$. **[2]**

(ii) Hence find the maximum speed, giving your answer in terms of a and g. **[4]**

(c) In reality, the spring is not light.
State one way in which this would affect your energy equation in part (b). **[1]**

9. A light elastic string with natural length l and modulus of elasticity kmg has one end attached to a fixed point A on a rough inclined plane. The other end of the string is attached to a package of mass m. The plane is inclined at an angle θ to the horizontal, where

$$\tan\theta = \frac{3}{4}$$

The package is initially held at the point C on a line of greatest slope of the plane below A, where $AC = 3l$. The package is then projected down the plane with speed $2\sqrt{gl}$ and first comes to rest at the point B, where $AB = 5l$.

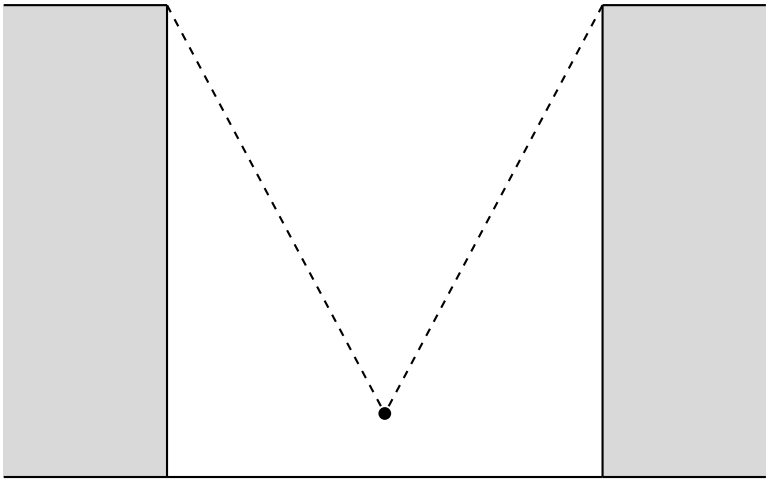
The coefficient of friction between the package and the plane is $\frac{1}{2}$.

By modelling the package as a particle,

(a) show that

$$k = \frac{2}{5}$$ **[6]**

(b) find the acceleration of the package at the instant it starts to move back up the plane from the point B. **[5]**



10. A stunt performer is attached to two identical light elastic cords. One end of each cord is fixed to opposite sides of the rim of a vertical shaft, as shown in the diagram above.

The performer has mass 80 kg and is modelled as a particle. Initially the performer is held at rest at the centre of the shaft, 7 m above the floor. The shaft is 52 m deep and 48 m wide. Each elastic cord has natural length 30 m and modulus of elasticity 3150 N.

The performer is released from rest.

(a) Show that the speed of the performer when the cords first become slack is $25\,\mathrm{m\,s^{-1}}$ correct to two significant figures. **[6]**

(b) Determine whether the performer is moving up or down when the cords become taut again. **[5]**